RESEARCH / COMPUTATIONAL SCIENCE

MODELING / LEARNING / SIMULATION

Reego John-Annaselvan

Computer Science / Modeling / Machine learning

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A / EDUCATION

Georgia Institute of Technology / BS Computer Science / Graduation Spring 2027 / **GPA 3.54/4.0** / Threads: Modeling & Simulation; Intelligence / **The College of New Jersey** / First-year Study

B / RESEARCH _ EXPERIENCE

Compound Semiconductor Products VIP

Subteam Lead / Georgia Tech / Dr. William Hunt
Advisor: bill.hunt@ece.gatech.edu

GaN strain, polarization, and atomic recoil modeling for neutron-detector research

- Lead the simulation subteam within Georgia Tech's VIP partnership with Sandia National Laboratories.
- Contribute to Quantum ESPRESSO strain-polarization calculations and LAMMPS recoil simulations.
- Assess numerical convergence, local strain, and the physical limits of detector models.

Stopping criteria for active learning

The College of New Jersey / Dr. Michael Bloodgood
Advisor: mbloodgood@tcnj.edu

Research on active-learning stopping criteria that balance annotation cost and predictive performance

- Researched active-learning stopping criteria with Dr. Michael Bloodgood using small-text.
- Studied when to stop requesting annotations while retaining predictive performance.

Nano PV Solar

Internship / Team leadership / Three years
Company contact: info@nano-pv.com

Computational characterization of photovoltaic device performance

- Led collaborative photovoltaic analysis and developed a Python framework for current-voltage characterization, device-parameter extraction, and model fitting.
- Evaluated parameter sensitivity and measurement variability across test conditions to distinguish device trends from testing artifacts.

RESEARCH + PROJECT TECHNOLOGIES

Languages: Python / C++ / SQL / MATLAB
Machine learning: PyTorch / JAX / small-text / scikit-learn
Scientific computing: NumPy / SciPy / Matplotlib
Physical simulation: Geant4 / Quantum ESPRESSO / LAMMPS
Graphics + parallel computing: CUDA / OpenGL / Slurm
Development + infrastructure: Linux / Git / Docker / CMake / pytest / GitHub Actions / Cloud computing

C / PERSONAL PROJECTS

Last Label / When should learning stop?

Python / small-text / scikit-learn

Annotation cost and predictive performance under noisy labels and biased stop sets

- Evaluated published SP and PCF stopping rules across 15,360 learning trajectories on six binary tasks.
- Analyzed paired cost-performance differences, budget fallbacks, noise interactions, and stopping-convention sensitivity.

SENSORLAB / Learning where to measure

Thermal inference / Experimental design

Simulation guided sensor placement for thermal property estimation.

Built a differentiable heat solver to estimate material properties from sparse, noisy measurements.

Validated gradients and compared sensor layouts using equal measurement budgets.

LUMEN / CPU and GPU path tracer

C++ / CUDA / OpenGL

Path tracing for physically based lighting simulation

- Built a multithreaded CPU path tracer with BVH traversal, area lights, and multiple importance sampling.
- Implemented a CUDA wavefront backend and progressive OpenGL viewer.

RESEARCH INTERESTS

Apply machine learning and simulation to practical engineering problems. Improve computational efficiency and guide useful experiments.

MODEL / MEASURE / EXPLAIN: Connecting physical simulation, data-efficient learning, and computational performance. Showcased on [rja.dev] / [reego-john-annaselvan.com]